

Case Study Rubric

Due: TBD

Submission format: Upload pdf and link to github repo to Canvas

General Description

Submit to Canvas your written analysis PDF and link to your GitHub repository. Your deliverable should demonstrate your ability to combine technical and conceptual skills in a hands-on scenario similar to what you may encounter in academic courses or professional environments.

Why am I doing this?

This case study is an opportunity to synthesize technical and conceptual skills developed throughout this course. You will apply statistical analysis, data visualization, and critical thinking to investigate a real-world question: whether fantasy football media sentiment provides genuine predictive signal for NFL player performance.

What am I going to do?

You will produce a comprehensive report that investigates sentiment analysis applied to fantasy football expert articles. Your analysis will determine whether analyst tone genuinely predicts player performance or simply recycles existing statistics, biases, and media narratives. The deliverable will include:

- Written Analysis PDF - A detailed report presenting your findings, visualizations, and conclusions
- Exploratory data analysis with clear visualizations
- Statistical modeling and hypothesis testing
- Bias detection and critical analysis
- Code & Data - All Python code and notebooks created for your analysis in a GitHub repository
- Code should be reproducible and well-documented
- Include any data transformations or preprocessing
- References - References and citations on a separate page at the end of the written analysis PDF using IEEE citation style

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Tips for Success

- Plan your analysis before coding: sketch the EDA steps, models to test, and key plots you want to include.
- Reproduce the Technical Reading examples first, then extend them with new checks and robustness tests.
- Use the player-week combination as your primary unit of analysis to avoid duplicate observations.

- Use version control: commit notebooks and scripts with clear messages; host everything in the required GitHub repo.
- Proofread the written PDF and have a peer review it before submission.

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How will I know I have succeeded?

You will meet expectations on this case study when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	• Written Analysis ○ Submit the analysis as a PDF file ○ Include exploratory plots, model outputs, and visualizations throughout • Code & Data ○ Submit all analysis code in a GitHub repository ■ Include source code, README.md, LISCENCE.md, data, and references. ○ Include all data used or generated during analysis ○ Document code with clear comments and markdown explanations
Written Analysis	Detailed expectations for the PDF submission: • Structure: Title page, one-paragraph Hook, Methods, Results, Conclusion, References, Appendix (code links). • Content: Clear problem statement, motivation, EDA, modeling approach, bias checks, and final recommendations. • Length & Format: 4-6 pages (excluding appendix), PDF format. Figures and tables should be embedded and legible.
Code & Technical Material	Description of the Github repository contents: • Notebooks/Scripts: All analysis code (Jupyter notebooks or Python scripts) with clear markdown and comments. • Data: Include small/cleaned datasets or provide links to large data stored elsewhere. Document data sources and any preprocessing. • Environment: requirements.txt and brief README reproduction steps.
Rubric	• Project purpose and quick summary of findings. • How to run the notebooks/scripts to reproduce results (commands). • File/folder structure and where to find key outputs. • Link to the PDF submission and citation information.

References	● Include a reference page in the PDF using IEEE citation style. ● Cite datasets, sentiment-analysis tools/libraries, and any articles used as supporting materials. ● Include links or DOIs where available.
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